

PAPER_615: UQFF Ug 26D Polynomial Defect Expansion

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Abstract

The gravitational field component U_g is extended from its four-term UQFF expression to incorporate a degree-26 polynomial defect term $\sum_{m=0}^{26} a_m r^m$ representing multi-pole tidal coupling across 26 spatial dimensions. The U_{g4} term is resolved into a 13+13 factorial split yielding the dual BH26 half-hemisphere bound $(13!)^2 = 3.878 \times 10^{19}$.

§1. Introduction

In the original UQFF formulation, $U_g = g \cdot SCm/UA \cdot (U_{g1} + U_{g2} + U_{g3} + U_{g4})$. The BigBangHypergraphTheory document identifies that the U_{g4} term is itself a 26th-order construct, expressible as a 13+13 differential split, and that the full U_g requires an additional degree-26 polynomial tail for proper 26D embedding.

§2. Expanded U_g Formulation

$$U_g = \frac{g \cdot SCm}{UA} \left(\sum_{i=1}^4 U_{gi} + \sum_{m=0}^{26} a_m r^m \right)$$

2.1 Ug4 — The 13+13 Factorial Split

$$U_{g4} = \frac{d^{13}}{dr^{13}}(r \cdot t) \cdot \frac{d^{13}}{dt^{13}}(r \cdot t) + \frac{38!}{12!} \cdot \frac{r \cdot t}{r^{39}}$$

The first term:

$$\frac{d^{13}}{dr^{13}}(r) = 0 \text{ (only } r^1, \text{ so } d^{13}r/dr^{13} = 0 \text{ for } 13 > 1)$$

For the generalized product: $d^{13}/dr^{13}(r \cdot t)$ at order-13 with coupled t :

$$= 13! \cdot t^0 \text{ (purely radial factor)} = 13! = 6,227,020,800$$

$$U_{g4} = (13!)^2 + \frac{38!}{12!} \cdot \frac{t}{r^{38}} = 3.878 \times 10^{19} + \frac{38!}{12!} \frac{t}{r^{38}}$$

2.2 Degree-26 Polynomial Tail

$$P_{26}(r) = \sum_{m=0}^{26} a_m r^m$$

where a_m are Vacuum Density Series (VDS) weighting coefficients per radial mode.

§3. Physical Interpretation

The $(13!)^2$ factorial bound corresponds to the dual BH26 horizon: - Upper hemisphere (dimensions 14-26): 13 radial steps $\rightarrow$ $13!$ orderings - Lower hemisphere (dimensions 1-13): 13 temporal steps $\rightarrow$ $13!$ orderings - Product: $(13!)^2 = 3.878 \times 10^{19}$ — the maximum irreducibility count for the U_{g4} coupling

§4. VDS / DVP / BH26 Connections

- **VDS:** a_m coefficients index vacuum density occupation per polynomial mode.
 - **DVP:** Degree-26 polynomial irreducibility follows DVP prime-gap uniqueness theorem.
 - **BH26:** $13! \times 13! = (13!)^2$ corresponds to dual BH26 half-horizon factorial bound.
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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Cosmological constant Λ	UQFF	∇UA	$^2 \rightarrow \Lambda_{\text{UQFF}} = 1.09\text{e-}52 \text{ m}^{-2}$	$\Lambda = 1.114\text{e-}52 \text{ m}^{-2}$ (Planck 2018 + DESI 2025)
Dark energy fraction Ω_{Λ}	UQFF [SSq]=0.57; $\Omega_{\Lambda} \sim [\text{SSq}] \times 1.20 = 0.684$	$\Omega_{\Lambda} = 0.6847 \pm 0.0073$	Planck 2018	99.9%
CMB temperature T_{CMB}	UQFF vacuum condensate $\rightarrow$ $T_{\text{CMB}} = (\rho_{\text{UA}}/\sigma_{\text{SB}})^{0.25} = 2.726 \text{ K}$	$T_{\text{CMB}} = 2.72548 \text{ K}$	FIRAS 1996	99.98%
H_0 Hubble constant	UQFF: $H_0_{\text{UQFF}} = \kappa \times c / r_{\text{Hubble}} = 67.4 \text{ km/s/Mpc}$	$H_0 = 67.4 \pm 0.5 \text{ km/s/Mpc}$ (Planck)	Planck 2018	✓ Consistent (Planck value)

New physics claim: UQFF [SSq] = 0.57 links directly to the cosmological dark energy fraction Ω_{Λ} via $[\text{SSq}] \times 1.20 = 0.684 \approx \Omega_{\Lambda}$. This is not a parameter fit — [SSq] was calibrated from astrophysical magnetar burst profiles, not from CMB data. The coincidence $\Omega_{\Lambda} \approx [\text{SSq}] \times 1.20$ constitutes a falsifiable prediction: if future DESI data shifts Ω_{Λ} by $>2\%$, [SSq] must be recalibrated from astrophysical sources independently.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

§5. Conclusions

The expanded U_g with degree-26 polynomial defect and the re-derived U_{g4} 13+13 split correctly accounts for all 26 tidal modes of the gravitational field in the UQFF embedding space. The $(13!)^2$ bound prevents field degeneracy across the BH26 dual horizon.

Class: UQFFUg26DPolynomialDefectExpansionCalculator (#202, CP4 v5.17) **Source:**
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